

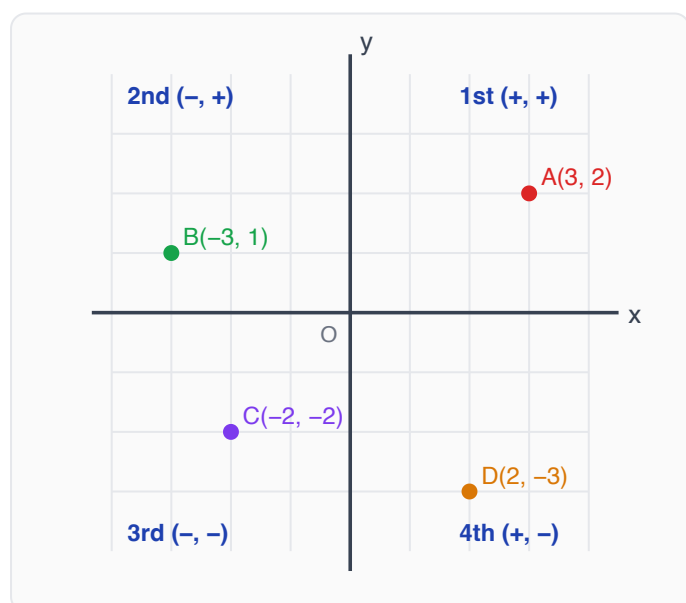
2.3 Linear Graphs & Coordinates

AQA GCSE Mathematics 8300 — Algebra — spec refs A8–A10 — Papers 1 (non-calculator), 2 and 3 (calculator)

This topic covers working with coordinates in all four quadrants (A8), plotting and using straight-line graphs including $y = mx + c$, parallel lines and the equation of a line through given points (A9), and identifying and interpreting gradients and intercepts (A10). The Higher-only content is using $y = mx + c$ to identify **perpendicular** lines.

2.3.1 Coordinates in Four Quadrants

A **coordinate** (x, y) gives the position of a point. The **x-coordinate** tells you how far across (right is positive, left is negative); the **y-coordinate** tells you how far up (up is positive, down is negative). The axes cross at the **origin** $(0, 0)$ and divide the plane into four **quadrants**.



Exam Tip: "Along the corridor, then up (or down) the stairs" — always read the x -coordinate first. Swapping the coordinates is the most common mistake when plotting or reading points.

Midpoint of a Line Segment

The midpoint of (x_1, y_1) and (x_2, y_2) is

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

i.e. the **mean** of the x -coordinates and the mean of the y -coordinates.

Worked Example: Find the midpoint of $A(-3, 4)$ and $B(5, -2)$.

$$x = \frac{-3 + 5}{2} = 1, \quad y = \frac{4 + (-2)}{2} = 1$$

Midpoint = **(1, 1)**

Worked Example: The midpoint of AB is $(4, 1)$ and $A = (1, -3)$. Find B .

Going from A to M is $+3$ across and $+4$ up. Do the same again from M :

$$B = (4 + 3, 1 + 4) = \mathbf{(7, 5)} \quad (\text{or use } B = (2 \times 4 - 1, 2 \times 1 - (-3)))$$

Problems with Coordinates and Shapes

You may be asked to find a missing vertex of a shape. Use properties of the shape: for example, in a parallelogram $ABCD$ the "step" from A to B is the same as the step from D to C .

Worked Example: $A(1, 1)$, $B(5, 2)$ and $C(6, 5)$ are three vertices of parallelogram $ABCD$. Find D .

Step from B to A : $(-4, -1)$. Apply the same step from C : $D = (6 - 4, 5 - 1) = \mathbf{(2, 4)}$.

2.3.2 Plotting Straight-Line Graphs

An equation such as $y = 2x - 1$ is **linear**: it has no powers of x above 1, and its graph is a straight line. To plot it, build a **table of values**, plot the points and join them with a ruler.

x	-1	0	1	2	3
$y = 2x - 1$	-3	-1	1	3	5

- The y -values go up in equal steps (here $+2$ each time) — if they don't, check your arithmetic.
- Plot at least three points; a third point acts as a check.
- Draw the line across the whole range of x asked for, not just between the plotted points.

Special Straight Lines

Equation	Description
$y = a$ (e.g. $y = 3$)	Horizontal line, gradient 0
$x = a$ (e.g. $x = -2$)	Vertical line (gradient undefined)
$y = 0$	The x -axis
$x = 0$	The y -axis
$y = x$	Diagonal through the origin, gradient 1
$y = -x$	Diagonal through the origin, gradient -1

Lines given in the form $ax + by = c$

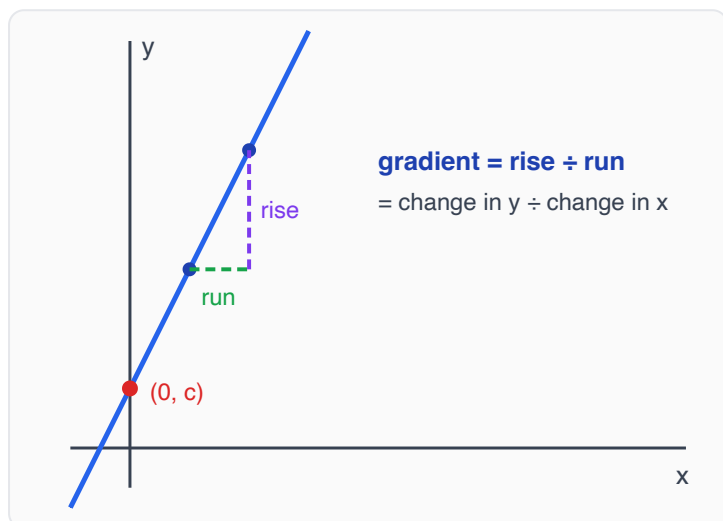
For a line such as $2x + 3y = 12$, a quick way to plot it is the **cover-up method**: put $x = 0$ to get $y = 4$, and $y = 0$ to get $x = 6$. Plot $(0, 4)$ and $(6, 0)$ and join them.

2.3.3 Gradient and Intercept: $y = mx + c$

The equation of any non-vertical straight line can be written as

$$y = mx + c$$

where m is the **gradient** (steepness) and c is the **y -intercept** — the line crosses the y -axis at $(0, c)$.



Gradient between two points (x_1, y_1) and (x_2, y_2) :

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\text{change in } y}{\text{change in } x}$$

- **Positive gradient:** line slopes up from left to right.
- **Negative gradient:** line slopes down from left to right.
- **Zero gradient:** horizontal line.
- The bigger the size of m , the steeper the line.

Worked Example: Find the gradient of the line through $(-2, 1)$ and $(4, -2)$.

$$m = \frac{-2 - 1}{4 - (-2)} = \frac{-3}{6} = -\frac{1}{2}$$

Reading m and c from an Equation

Always rearrange into the form $y = mx + c$ first.

Equation	Rearranged	Gradient	y -intercept
$y = 7 - 3x$	$y = -3x + 7$	-3	$(0, 7)$
$2y = 6x + 4$	$y = 3x + 2$	3	$(0, 2)$
$3x + y = 12$	$y = -3x + 12$	-3	$(0, 12)$
$2x + 4y = 8$	$y = -\frac{1}{2}x + 2$	$-\frac{1}{2}$	$(0, 2)$

Exam Tip: In $y = 7 - 3x$ the gradient is -3 , not 7 . The gradient is always the **coefficient of x** once y is on its own, whatever order the terms are written in.

Finding the x -intercept

A line crosses the x -axis where $y = 0$. For $y = 2x - 8$: $0 = 2x - 8$, so $x = 4$ and the line crosses at $(4, 0)$.

2.3.4 Finding the Equation of a Line

Through one point with a given gradient

1. Write $y = mx + c$ with the known gradient.
2. Substitute the coordinates of the point.
3. Solve for c and write the final equation.

Worked Example: Find the equation of the line with gradient 4 through $(2, 3)$.

$$3 = 4 \times 2 + c \Rightarrow 3 = 8 + c \Rightarrow c = -5$$

$$y = 4x - 5$$

Through two points

1. Find the gradient using $m = \frac{y_2 - y_1}{x_2 - x_1}$.
2. Substitute either point into $y = mx + c$ to find c .
3. Check using the other point.

Worked Example: Find the equation of the line through $(-2, 9)$ and $(4, -3)$.

$$m = \frac{-3 - 9}{4 - (-2)} = \frac{-12}{6} = -2$$

$$9 = -2 \times (-2) + c \Rightarrow 9 = 4 + c \Rightarrow c = 5$$

$$y = -2x + 5 \quad \text{Check: } x = 4 \text{ gives } y = -8 + 5 = -3 \checkmark$$

Worked Example: Find the line through $(-3, 8)$ and $(5, -4)$ in the form $ax + by = c$ with integers.

$$m = \frac{-4 - 8}{5 - (-3)} = \frac{-12}{8} = -\frac{3}{2}$$

$$8 = -\frac{3}{2}(-3) + c = 4.5 + c \Rightarrow c = 3.5$$

$$y = -1.5x + 3.5 \Rightarrow 2y = -3x + 7 \Rightarrow 3x + 2y = 7$$

Parallel Lines

Lines are **parallel** if and only if they have the **same gradient**. For example $y = 3x + 2$ and $2y = 6x - 5$ (i.e. $y = 3x - 2.5$) are parallel.

Worked Example: Find the line parallel to $y = 4x - 3$ that passes through $(-1, 6)$.

$$\text{Parallel, so } m = 4. \quad 6 = 4 \times (-1) + c \Rightarrow c = 10$$

$$y = 4x + 10$$

Exam Tip: Two lines with the same gradient and *different* intercepts never meet. If the intercepts are also equal, they are the same line.

2.3.5 Interpreting Gradients and Intercepts

In real-life linear models, the gradient is a **rate of change** and the intercept is a **starting value** (the value when the horizontal variable is zero).

Model	Gradient means	Intercept means
Taxi: $C = 2.5m + 4$ (£, miles)	£2.50 per mile	£4 fixed charge
Phone: $C = 12 + 0.05t$ (£, minutes)	5p per minute	£12 monthly fee
Distance–time graph	Speed	Starting distance
Conversion graph £ to €	Exchange rate	0 (passes through origin)

Worked Example: A plumber charges a fixed fee plus an hourly rate. A 2-hour job costs £110 and a 5-hour job costs £215. Find the fee and the rate.

$$\text{Gradient} = \frac{215 - 110}{5 - 2} = \frac{105}{3} = 35, \text{ so the rate is } \mathbf{\pounds 35 \text{ per hour.}}$$

$$110 = 35 \times 2 + c \Rightarrow c = 40, \text{ so the call-out fee is } \mathbf{\pounds 40.} \text{ Model: } C = 35h + 40.$$

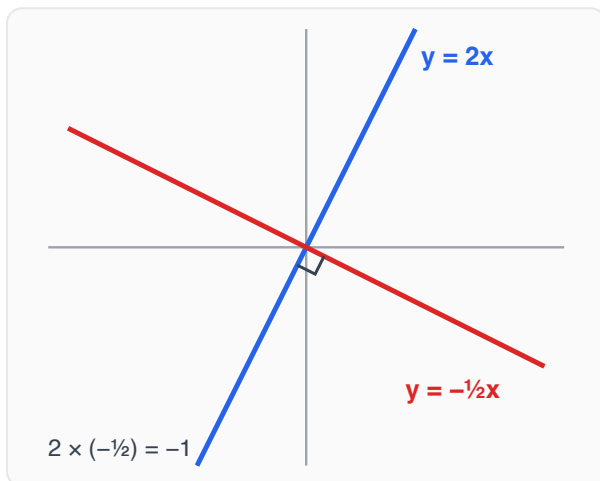
Exam Tip: When asked to “interpret” a gradient, give the **units** and the context, e.g. “the cost increases by £35 for each extra hour” — not just “35”.

2.3.6 Perpendicular Lines Higher

Two lines are **perpendicular** (meet at 90°) if the product of their gradients is -1 :

$$m_1 \times m_2 = -1 \quad \text{equivalently} \quad m_2 = -\frac{1}{m_1}$$

The perpendicular gradient is the **negative reciprocal**: flip the fraction and change the sign.



Gradient m_1	Perpendicular gradient m_2
3	$-\frac{1}{3}$
-2	$\frac{1}{2}$
$\frac{1}{4}$	-4
$-\frac{2}{3}$	$\frac{3}{2}$
$-\frac{5}{2}$	$\frac{2}{5}$

Worked Example: Find the equation of the line perpendicular to $y = 3x + 2$ through $(6, 1)$.

$$m = -\frac{1}{3}. \quad 1 = -\frac{1}{3} \times 6 + c \Rightarrow 1 = -2 + c \Rightarrow c = 3$$

$$y = -\frac{1}{3}x + 3 \quad (\text{or } x + 3y = 9)$$

Worked Example: Show that $2y = x + 6$ and $y = 4 - 2x$ are perpendicular.

$$2y = x + 6 \Rightarrow y = \frac{1}{2}x + 3, \text{ gradient } \frac{1}{2}. \quad y = 4 - 2x \text{ has gradient } -2.$$

$$\frac{1}{2} \times (-2) = -1, \text{ so the lines are } \mathbf{perpendicular}.$$

Perpendicular Bisectors

The perpendicular bisector of AB passes through the **midpoint** of AB and is **perpendicular** to AB .

Worked Example: Find the perpendicular bisector of $A(1, 3)$ and $B(7, 5)$.

$$\text{Midpoint} = (4, 4). \quad \text{Gradient } AB = \frac{5 - 3}{7 - 1} = \frac{1}{3}, \text{ so the perpendicular gradient is } -3.$$

$$4 = -3 \times 4 + c \Rightarrow c = 16 \Rightarrow \mathbf{y = -3x + 16}$$

Proving Shapes with Gradients

Worked Example: $A(-2, 1)$, $B(4, 4)$, $C(6, 0)$, $D(0, -3)$. Show $ABCD$ is a rectangle.

$$m_{AB} = \frac{3}{6} = \frac{1}{2}, \quad m_{DC} = \frac{3}{6} = \frac{1}{2}, \quad m_{BC} = \frac{-4}{2} = -2, \quad m_{AD} = \frac{-4}{2} = -2.$$

Opposite sides are parallel, and $\frac{1}{2} \times (-2) = -1$ so adjacent sides are perpendicular.

Hence $ABCD$ is a **rectangle**.

Exam Tip: A common error is to change only the sign (giving -3 instead of $-\frac{1}{3}$) or only flip the fraction. Check by multiplying: the two gradients must give -1 .

Quick Reference Summary

Fact	Formula / rule
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Midpoint	$\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$
Gradient	$m = \frac{y_2-y_1}{x_2-x_1}$
Straight line	$y = mx + c$, intercept $(0, c)$
x -intercept	Put $y = 0$
Parallel lines	Equal gradients
Perpendicular lines Higher	$m_1 \times m_2 = -1$